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Clinical Characteristics of 60 COVID-19-infected Patients with or without Renal Injury in Hangzhou, China --Manuscript Draft--

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Abstract:	Summary Background Up to now, COVID-19 has infected more than 60 thousand of people in China and is gradually spread globally. The disease was named as Novel coronavirus pneumonia (NCP) on February 8th. We analyzed data on 60 confirmed cases in Hangzhou to determine the clinical characteristics of NCP with or without renal injury. Methods In this cross-sectional, descriptive study, we collected clinical information on demographic characteristics, pulmonary high resolution computerized tomography(HRCT) imaging and other clinical indicators of NCP cases. Urine protein(Upro) was used to divided patients into the Upro positive group (Upro+ group) and the negative Upro group(Upro- group). Then difference of clinical indexes between the two groups and relationship between Upro and clinical indexes were analyzed. Findings Among the 60 patients, Upro positive rate was 51.67%, and urine protein and urine erythrocyte co-positive rate was 11.67%. The level of blood lymphocyte, serum Age, married, BMI, history of hypertension, ALT, AST, Glb, Scr, LDH, lymphocyte, CRP and moderate pulmonary HRCT imaging of Upro+ group were

	remarkably severe than those of Upro- group ($P < 0.05$). Abnormal rates of AST and CRP were significantly higher in Upro+ group. The results of binary regression analysis showed that Scr(OR=1.1;), AST(OR=1.199), age(OR=1.1) and grade of pulmonary HRCT imaging(OR=0.315) were homodirectional impact factors and blood lymphocyte(OR=0.222) was the reverse impact factor of urine protein positive. Interpretation High rate of Upro positive was detected in NCP patients, The degree of pulmonary lesion is positively related with level of Upro. Moreover, age, AST and lymphopenia were homodirectional impact factors with Upro. This study reminded us to pay more attention to the protective measures for lung, kidney and heart in the early stage of clinical treatment, and the mechanisms of how COVID-19 induced renal injury need to be future studied . Funding Zhejiang Natural Science Foundation Committee
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26 **Summary**

27 **Background** Up to now, COVID-19 has infected more than 60 thousand of people in China and is
28 gradually spread globally. The disease was named as Novel coronavirus pneumonia (NCP) on February
29 8th. We analyzed data on 60 confirmed cases in Hangzhou to determine the clinical characteristics of
30 NCP with or without renal injury.

31 **Methods** In this cross-sectional, descriptive study, we collected clinical information on demographic
32 characteristics, pulmonary high resolution computerized tomography(HRCT) imaging and other
33 clinical indicators of NCP cases. Urine protein(Upro) was used to divided patients into the Upro
34 positive group (Upro+ group) and the negative Upro group(Upro- group). Then difference of clinical
35 indexes between the two groups and relationship between Upro and clinical indexes were analyzed.

36 **Findings** Among the 60 patients, Upro positive rate was 51.67%, and urine protein and urine
37 erythrocyte co-positive rate was 11.67%. The level of blood lymphocyte, serum Age, married, BMI,
38 history of hypertension, ALT, AST, Glb, Scr, LDH, lymphocyte, CRP and moderate pulmonary
39 HRCT imaging of Upro+ group were remarkably severe than those of Upro- group ($P < 0.05$).
40 Abnormal rates of AST and CRP were significantly higher in Upro+ group. The results of binary
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42 HRCT imaging(OR=0.315) were homodirectional impact factors and blood lymphocyte(OR=0.222)
43 was the reverse impact factor of urine protein positive.

44 **Interpretation** High rate of Upro positive was detected in NCP patients, The degree of pulmonary
45 lesion is positively related with level of Upro. Moreover, age, AST and lymphopenia were
46 homodirectional impact factors with Upro. This study reminded us to pay more attention to the
47 protective measures for lung, kidney and heart in the early stage of clinical treatment, and the
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Abbreviations

NCP	Novel coronavirus pneumonia
COVID-19	Coronavirus Disease 2019
COV	coronavirus
SARS	Severe acute respiratory syndrome
MERS	Middle East respiratory syndrome
ACE2	angiotensin converting enzyme II
Upro	Urine protein
URBC	Urine Red Blood Cell
BMI	Body Mass Index
ALT	Alanine Aminotransferase
AST	Aspartate Aminotransferase

Glb	Globulin
Alb	Albumin
Scr	Serum creatinine
BUN	Blood Urea Nitrogen
LDH	Lactate dehydrogenase
CK	Creatine Kinase
WBC	White Blood Cell
NEUT	Neutrophil
LY	lymphocyte
RBC	Red Blood Cell
HGB	Hemoglobin
CRP	C-reactive Protein
Ridit	Relative to an identified distribution unit

60 **Introduction**

61 COVID-19 was identified as causative pathogen of pneumonia which was lately named as Novel
62 coronavirus pneumonia (NCP). NCP rapidly spread from Wuhan to other areas of China and abroad.
63 On Feb 15, the National Health and Commission of People's Republic of China announced that there
64 were 66580 confirmed NCP cases and 1524 death in China. It has been incorporated into the category
65 B infectious diseases stipulated in the Law of the People's Republic of China on the Prevention and
66 Control of Infectious Diseases. COVID-19 is a novel beta-coronavirus belonging to the sarbecovirus
67 subgenus of Coronaviridae family, which is distinct from SARS-CoV and MERS-CoV¹ that
68 respectively associated with human severe acute respiratory syndrome (SARS)² and Middle East
69 respiratory syndrome (MERS)³, respectively.

70 A 41-NCP-cases research showed the common symptoms at onset of NCP were fever, cough, and
71 myalgia or fatigue⁴. In clinical practice, we found that there were a lot of NCP patients with renal
72 injury, and therefore, this paper is going to figure out the relationship between clinical characteristics of
73 the 60 NCP cases with renal injury (Upro positive) in Hangzhou, China.

74 **Methods**

75 **Patients**

76 60 NCP patients in Hangzhou Xixi Hospital affiliated to Zhejiang Chinese Medicine University were

selected by Feb 15, 2020. From Jan 22 to Feb 15, 2020, these patients were all diagnosed according to the Diagnosis and Treatment Scheme for New Coronavirus Pneumonia (Pilot Edition 5) issued by the National Health and Commission of People's Republic of China on Feb 4, 2020.

The diagnostic criteria is as follows: First, any one of the history of epidemiological, or any two of the clinical manifestations, can be diagnosed as a suspected case: (1) history of Epidemiology: I. travel history or residence history of Wuhan City and surrounding areas within 14 days prior to onset of the disease, or other cases reporting community; II. contact with patients from Wuhan City and surrounding areas, or with fever or respiratory symptoms from the community prior to onset of the disease, within 14 days prior to onset of the disease; III. cluster disease; and IV. contact with a new type of coronavirus infection (the new type of coronavirus infection refers to the positive detection of pathogenic nucleic acid); (2) clinical manifestations I. fever and/or respiratory symptoms; II. imaging features of the above pneumonia; III. normal or decreased total white blood cell count or decreased lymphocyte count at the early stage of onset. Second, suspected cases with one of the following etiological evidence diagnosed as NCP: (1) COVID-19 nuclear acid positive in respiratory or blood samples detected by real-Time reverse transcription polymerase chain reaction (RT-PCR); (2) genetic sequence of virus in respiratory or blood specimen is highly homologous with COVID-19 detected by genetic sequencing. This study was approved by the Ethics Commission of Hangzhou Xixi Hospital affiliated to Zhejiang Chinese Medical University (2020- KLS-15).

Procedures

A cross-sectional, descriptive study was carried out to analyze the clinical characteristics of these NCP cases. For this objective, the first clinical informations on demographic characteristics, clinical symptoms, imaging changes in chest high resolution computerized tomography(HRCT), Urine protein(Upro), Urine RBC and WBC, blood cell count, serum glutamate aminotransferase(ALT), glutamate aminotransferase (AST), creatinine(Scr) Urea nitrogen(BUN), c-reactive protein(CRP) of all the NCP cases without being treated with antiviral, analgesic-antipyretic and antibiotics drugs was collected.. Then the patients were divided into Upro positive group (Upro+ group) and Upro negative group(Upro- group). The difference of other clinical indexes between the two groups and the relativity between the clinical indexes and urine protein were analyzed by regression analysis.

Data collection

Clinical medical records, laboratory findings, reports and imaging of chest HRCT for all the diagnosed NCP patients were collected.. The epidemiological data was gathered by directly communication with patients or their families. After independently reviewed,, the data were entered into standardised data collection Electronic Clinical Report Form(eCRF) by two researchers, and the system would check the entered data.

Detection of COVID-19 in throat swab

Detection of COVID-19 in throat swab is the main diagnose method for NCP by RT-PCR assay. For extracting COVID-19 RNA, the throat swab samples were collected from patients suspected of having COVID-19 infection. In brief, the collected throat swabs were stored in the tubes with 150 μ L of

virus preservation solution for transfer to the laboratory, and total RNA was extracted within 2 hours using the respiratory sample RNA isolation kit (ZJ Bio-Tech. Shanghai, China) for RT-PCR assay of COVID-19 RNA. Three target genes, including RdRP gene(RdRP), nucleocapsid protein gene (N) and E protein gene(E) , were simultaneously amplified and tested during the real-time RT-PCR assay. Cycle threshold value (Ct-value) ≤ 43 was defined as a positive test result, and Ct-value >43 or not detected was defined as a negative test. Only if RdRP, N and E were all positive, or RdRP plus N or E were positive, or only RdRP repeated twice positive can be diagnosed as COVID-19 positive.

Definitions

After exclusion of drug-induced renal injury and history of nephropathy, renal injury was diagnosed based on Upro level. Meanwhile, exclusion covers UWBC positive patients. The imaging grade of chest HRCT was divided into 4 levels, the grading criteria as follows: severe(3+) :lobar consolidation; moderate(2+): grinding glass-like changes; mild(+): focal infiltration; normal(-): no infiltration.

Statistical analysis

Continuous variables were expressed as median (IQR) and compared with the independent-Samples T test or the Mann-Whitney U test; categorical variables were expressed as number (%) and compared by χ^2 test or Fisher's exact test between Upro+ group and Upro- group. The comparison of grade of HRCT imaging between Upro+ group and Upro- group were analyzed by Ridit analysis. Predictors of Upro positive were analyzed by logistic regression. A two-sided α of less than 0.05 was considered statistically significant. Statistical analyses were done using the SPSS for windows software, version 19.0, unless otherwise indicated.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Results

By Feb 15, 2020, there were total 60 NCP patients in Hangzhou Xixi Hospital affiliated to Zhejiang Chinese Medicine University, the designated hospital for NCP isolation treatment. All patients had data on urine test results. The urine test results showed 31 cases with Upro positive and 29 cases with Upro negative. Among the Upro positive patients, there were 19 cases with simple protein positive, 5 cases with Upro and URBC co-positive, 3 cases with Upro,URBC and UWBC co-positive, and 4 cases with Upro and UWBC co-positive (Table 1) .

To further analyze the demographic, clinical and radiographic status characteristics of the patients with urine protein positive, we divided the patients into Upro+ group and Upro- group. As shown in Table 2, there were significant difference in the age [49(35-57) vs 35(26-51) years, $P=0.002$], marriage situation [married, 50(96.77%) vs 30(72.41%) $P=0.008$], BMI [24(20.7-25.2) vs 21(20.3-24) Kg/m², $P=0.042$], history of hypertension[8(25.8%) vs 1(3.4%), $P=0.027$] and the level of ALT[21(15-35) vs 14(11-22) U/L, $P=0.048$], AST[27(23-40) vs 19(17-25) U/L, $P=0.001$], globulin (Glb) [32(30-35) vs

28(26-30) g/L, $P=0.001$], Scr[68(61-84) vs 62(53-72) $\mu\text{mol/L}$, $P=0.002$], serum CRP[17(8-28) vs 8(3-9) mg/L, $P=0.001$], blood lymphocyte $[1.1(0.8-1.2) \text{ vs } 1.3(1-1.9) \times 10^9/\text{L}$, $P=0.014$] and LDH [223 vs 159 U/L, $P=0.013$] between the Upro+ group and Upro- group. Values of LDH in 7 cases are higher than 470 U/L in Upro+ group versus 0 patient in Upro- group. And a patient, whose urine protein was severe positive(3+), had the highest CK value(999 U/L) and LDH value(630 U/L), and his pulmonary HRCT imaging was shown in Fig.1. Although there were no severe radiographic status, the rate of moderate grade of HRCT imaging change in Upro+ group was higher than that of Upro- group (61.29% vs 31.03%, $P=0.037$).

Furthermore, we compared the abnormal laboratory indicators between the two groups. The statistical analysis was shown in Table 3, that abnormal rates of AST and CRP of Upro+ group were significantly higher than those of Upro- group (29.03% vs 6.90% and 67.74% vs 37.93%, respectively).

Because the data of imaging changes in Pulmonary HRCT were grading data, we compared the grade of HRCT imaging between Upro+ group and Upro- group with Redit analysis. As shown in Table 4, the Redit value of Upro+ group was significantly higher than that of Upro- group ($P=0.000$), which indicated that the radiographic status of Upro+ group was sever than that of Upro- group.

Finally, we evaluated the effect of various clinical indicators on urine protein positive with binary regression analysis. During the analysis, we used Upro positive or negative as dependent variables, used age, Scr, AST, lymphocyte CRP, GIB, BMI and grade of pulmonary HRCT imaging as independent variables, among these independent variables, GIB, BMI and grade of pulmonary HRCT imaging were used as dummy variable classification. The results showed that Scr($\text{OR}=1.1$; 95%CI, 1.005-1.024), AST($\text{OR}=1.199$; 95%CI, 1.041-1.381), age($\text{OR}=1.1$; 95%CI, 0.9545-1.089) and grade of pulmonary HRCT imaging($\text{OR}=0.315$; 95%CI, 0.021-4.809) were the homodirectional impact factors and blood lymphocyte ($\text{OR}=0.222$; 95%CI, 0.030-1.658) was the reverse impact factor of Upro positive (Fig.2).

Discussion

From December 2019 to Feb 2020, NCP has rapidly spread from Wuhan to other areas of China and it gradually spreads abroad. Up to Feb 15, there are 66580 confirmed NCP cases and 1524 death in China,. Among them ,166 cases was diagosed in Hangzhou, Zhejiang province. Here, We report a cohort of 60 NCP patients in Hangzhou Xixi Hospital affiliated to Zhejiang Chinese Medicine University. The 60 patients diagnosed by detection of COVID-19 in throat swab and chest HRCT. The 60 patients didn't receive antiviral, analgesic-antipyretic and antibiotics drugs treatment, which ensured no impact on clinical symptoms, testing indicators and imaging of HRCT.

Previous researches reported that complications included acute respiratory distress syndrome, lymphopenia and others⁴. Upro positive was found in 51.67% NCP patients with no obvious renal function loss. Why COVID-19 causes renal injury?

COVID-19 belongs to the sarbecovirus subgenus of Coronaviridae Family¹. As similar to SARS-CoV and MERS-CoV, COVID-19 belongs to the coronavirus of zoonotic patients^{1,5} and may also causes severe acute respiratory syndrome^{4 6}. SARS-CoV was the pathogen responsible for the severe acute

respiratory syndrome outbreaks in China in 2002 and 2003², and MERS-CoV caused the severe respiratory disease outbreaks in the Middle East in 2012³. For understanding the pathogenesis and treatment strategies for SARS, I Hamming, et al. investigated the localization of angiotensin converting enzyme II (ACE2) protein in various human organs and found ACE2 was abundantly present in several organs, including the epithelia of lung, small intestine, kidney, and so on^{7,8}. ACE2 is a receptor of SARS-CoV⁷. As previous studies had reported, most SARS-CoV infected patients developed lymphopaenia⁹, renal and heart injury^{10,11}.

It has been known that COVID-19, similar to SARS-CoV, invades human cells in lung and small intestine via the receptor ACE2^{7,12-14}. So it can be assumed that, COVID-19, like SARS-CoV, might cause kidney and heart injury¹⁵, which has been preliminarily confirmed by this study, i.e. there were 51.67% NCP patients exhibiting renal injury (Upro positive). Furthermore, the patient with Upro 3+ and highest CK value(999 U/L) and LDH value(630 U/L) indicated that the COVID-19 might induce lung, renal and heart co-injury, which reminded us to pay attention to the protective measures for lung, kidney and heart in the early stage of clinical treatment⁴.

The result of Rdit analysis indicated that the radiographic status of Upro⁺ group was sever than that of Upro⁻ group, which further confirmed the pathological status of lung and kidney combined lesions, though these patients did not develop to the level of multiple organ dysfunction syndrome (MODS). What's more, as shown in the results of binary regression analysis, AST(OR=1.199; 95%CI, 1.041-1.381) was the main homodirectional impact factors and blood lymphocyte(OR=0.222; 95%CI, 0.030-1.658) was the reverse impact factor of Upro positive. Similar to LDH, AST is also a common laboratory indicator of myocardial damage. These results indicated that COVID-19 may susceptibly invaded lung, kidney, heart and lymph nodes through ACE2, which was widespread distribution in these organs⁷.

AS we known, similar to SARS patients¹⁶, MERS-CoV infected patients¹⁷, the COVID-19⁴ infection would alsolead to the body immune cell response, reduce amounts of proinflammatory factors (eg, IL6, IFN γ , IL1 β , MCP1) and CRP increasing, and even cause inflammatory storms, further aggravated organ damage, including lung, kidney, heart, and so on. Therefore, it is still important to prevent effectively early.

COVID-19 ia pandemic epidemic disease and a threat to global health. Here we advocated early clinical measures to protect the patient's multiple organ function. However, more in-depth researches are needed to carry out to comprehensive control of hazards caused by COVID-19.

Contributors

ZX, JB, ZC, JH and CW had the idea for and designed the study and had full access to all data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. JB, SL,HC, JQ, JG, LM, ZL, YT, YZ and ZL contributed to collect the clinical data. ZX, LH, JS and CW contributed to writing of the report. ZX, SL, JH and CW contributed to critical revision of the report. ZX and HY contributed to the statistical analysis. All authors contributed to data acquisition, data analysis, or data interpretation, and reviewed and approved the final version.

Declaration of interests

All authors declare no competing interests.

Data sharing

The data that support the findings of this study are available from the corresponding author on reasonable request. Participant data without names and identifiers will be made available after approval from the corresponding author and National Health Commission. After publication of study findings, the data will be available for others to request. The research team will provide an email address for communication once the data are approved to be shared with others. The proposal with detailed description of study objectives and statistical analysis plan will be needed for evaluation of the reasonability to request for our data. The corresponding author and National Health Commission will make a decision based on these materials. Additional materials may also be required during the process.

Acknowledgments

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References

1. Zhu N, Zhang D, Wang W, et al. A Novel Coronavirus from Patients with Pneumonia in China, 2019. *The New England journal of medicine* 2020.
2. Ksiazek TG, Erdman D, Goldsmith CS, et al. A novel coronavirus associated with severe acute respiratory syndrome. *The New England journal of medicine* 2003; **348**(20): 1953-66.
3. de Groot RJ, Baker SC, Baric RS, et al. Middle East respiratory syndrome coronavirus (MERS-CoV): announcement of the Coronavirus Study Group. *Journal of virology* 2013; **87**(14): 7790-2.
4. Huang C, Wang Y, Li X, et al. Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China. *Lancet* 2020.
5. Cui J, Li F, Shi ZL. Origin and evolution of pathogenic coronaviruses. *Nature reviews Microbiology* 2019; **17**(3): 181-92.
6. Assiri A, Al-Tawfiq JA, Al-Rabeeh AA, et al. Epidemiological, demographic, and clinical characteristics of 47 cases of Middle East respiratory syndrome coronavirus disease from Saudi Arabia: a descriptive study. *The Lancet Infectious diseases* 2013; **13**(9): 752-61.
7. Hamming I, Timens W, Bulthuis ML, Lely AT, Navis G, van Goor H. Tissue distribution of ACE2 protein, the functional receptor for SARS coronavirus. A first step in understanding SARS pathogenesis. *The Journal of pathology* 2004; **203**(2): 631-7.
8. Ding Y, He L, Zhang Q, et al. Organ distribution of severe acute respiratory syndrome (SARS)

associated coronavirus (SARS-CoV) in SARS patients: implications for pathogenesis and virus transmission pathways. *The Journal of pathology* 2004; **203**(2): 622-30.

9. Wong RS, Wu A, To KF, et al. Haematological manifestations in patients with severe acute respiratory syndrome: retrospective analysis. *BMJ* 2003; **326**(7403): 1358-62.

10. Ding Y, Wang H, Shen H, et al. The clinical pathology of severe acute respiratory syndrome (SARS): a report from China. *The Journal of pathology* 2003; **200**(3): 282-9.

11. Chu KH, Tsang WK, Tang CS, et al. Acute renal impairment in coronavirus-associated severe acute respiratory syndrome. *Kidney international* 2005; **67**(2): 698-705.

12. Wu Y. Compensation of ACE2 Function for Possible Clinical Management of 2019-nCoV-Induced Acute Lung Injury. *Virologica Sinica* 2020.

13. Peiris JS, Chu CM, Cheng VC, et al. Clinical progression and viral load in a community outbreak of coronavirus-associated SARS pneumonia: a prospective study. *Lancet* 2003; **361**(9371): 1767-72.

14. Leung WK, To KF, Chan PK, et al. Enteric involvement of severe acute respiratory syndrome-associated coronavirus infection. *Gastroenterology* 2003; **125**(4): 1011-7.

15. Tang JW, To KF, Lo AW, Sung JJ, Ng HK, Chan PK. Quantitative temporal-spatial distribution of severe acute respiratory syndrome-associated coronavirus (SARS-CoV) in post-mortem tissues. *Journal of medical virology* 2007; **79**(9): 1245-53.

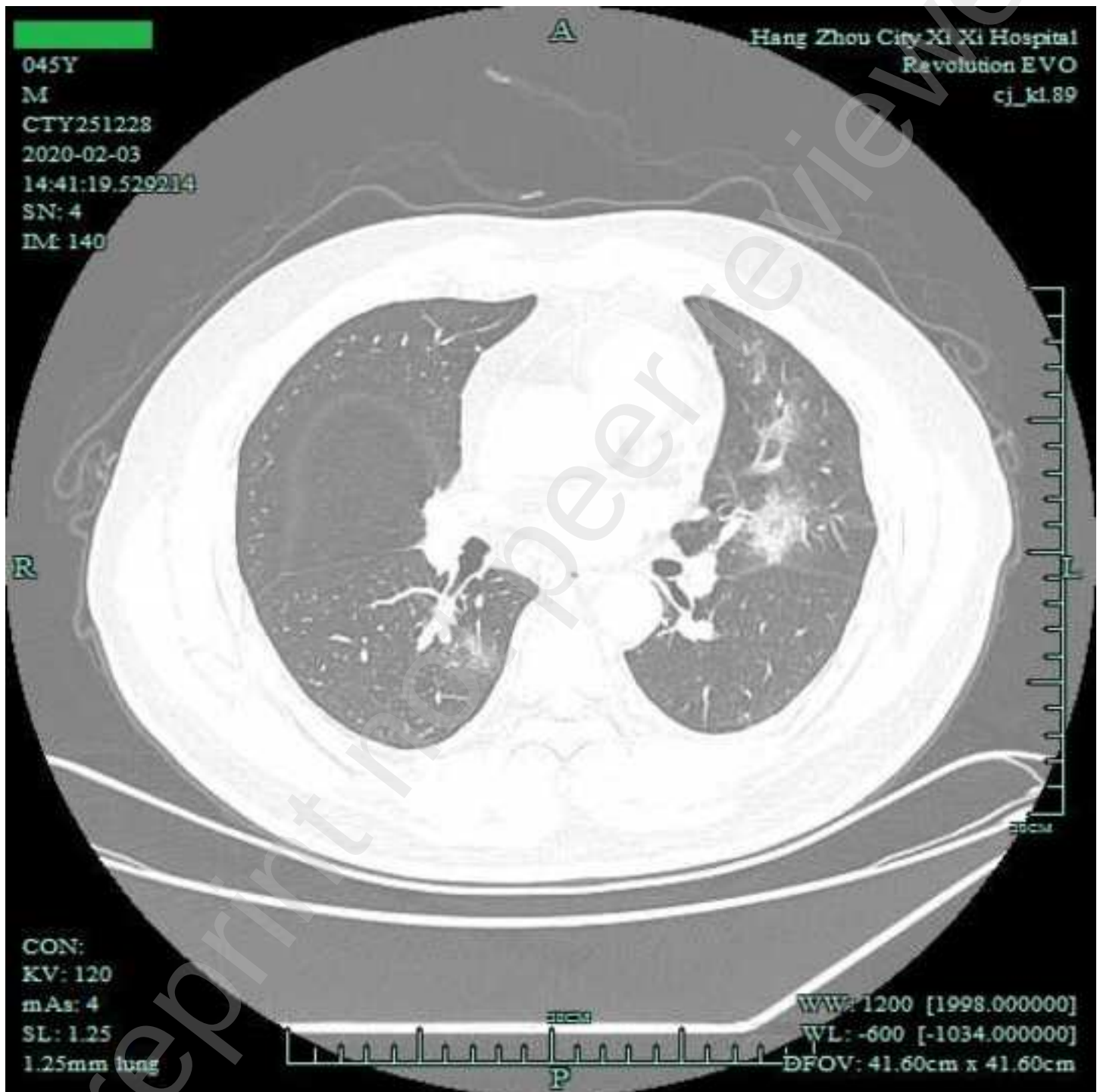
16. Wong CK, Lam CW, Wu AK, et al. Plasma inflammatory cytokines and chemokines in severe acute respiratory syndrome. *Clinical and experimental immunology* 2004; **136**(1): 95-103.

17. Mahallawi WH, Khabour OF, Zhang Q, Makhdoum HM, Suliman BA. MERS-CoV infection in humans is associated with a pro-inflammatory Th1 and Th17 cytokine profile. *Cytokine* 2018; **104**: 8-13.

Figures Legends

Fig. 1 Pulmonary HRCT imaging The pulmonary HRCT imaging of a urine protein severe positive patient, whose urine protein was severe positive(3+) and had the highest CK value(999 U/L) and LDH value(630 U/L) in all the 60 patients.

Fig. 2 Predictors of Upro positive by multivariate analysis (logistic regression). Independent variables listed (i.e. lymphocyte, AST, Scr, pulmonary HRCT Imaging, Age) were significant at $p < 0.05$, Upro⁺ group versus Upro⁻ group. OR, odds ratio.



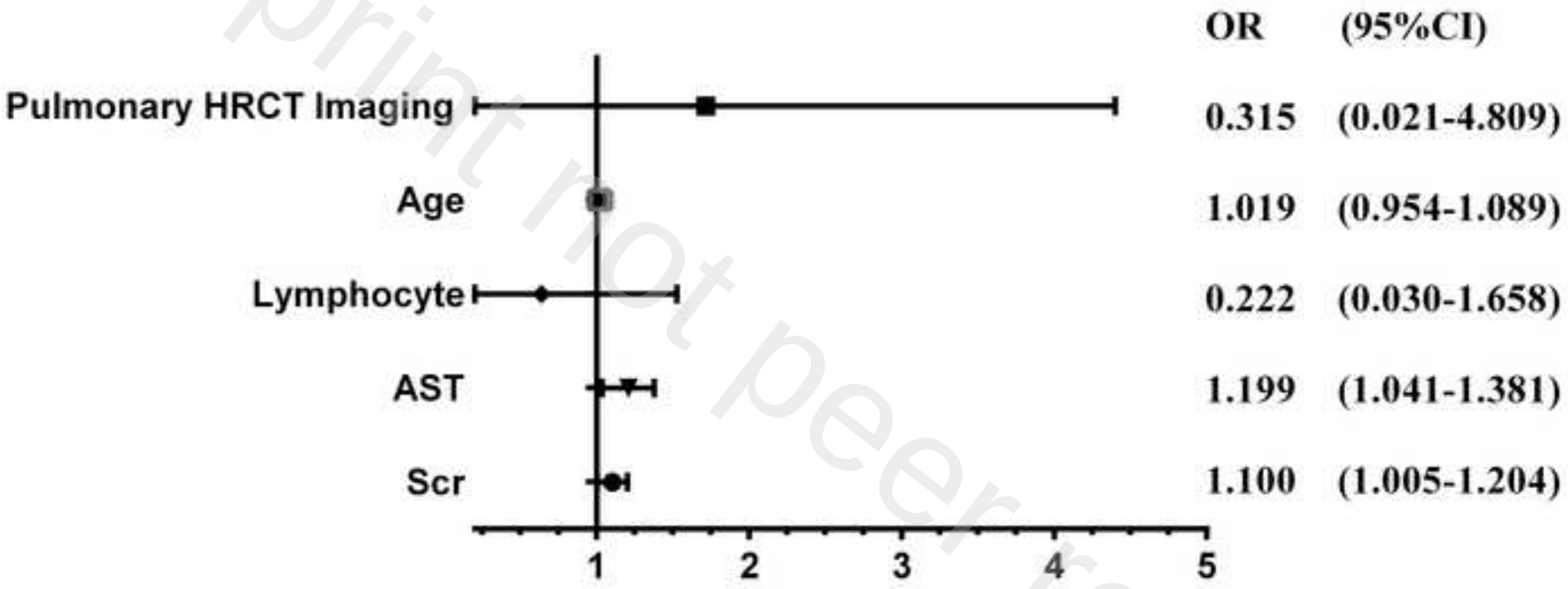


Table 1 Comparison of abnormal laboratory indicators between NCP patients with and without Upro positive [n(%)]

Parameter	Positive ^a				Negative
	Total	3+	2+	1+	
Urine protein	31(51.67)	1(1.67)	10(16.67)	20(33.33)	29(48.33)
Urine RBC	9(15)	1(1.67)	4(6.67)	4(6.67)	51(85.00)
Urine WBC	13(21.67)	5(8.33)	4(6.67)	4(6.67)	47(78.33)
Urine protein+RBC	5(6.67)	-	-	-	-
Urine protein+WBC	4(6.67)	-	-	-	-
Urineprotein+RBC+WBC	3(5.00)	-	-	-	-

Note. ^aQualitative examination of urinary protein: Negative (-): < 0.1 g/L, (+): 0.2-1.0g/L; (2+): 1.0-2.0 g/L, (3+):>2.0 g/L. Qualitative examination of urine RBC: Negative (-): < 6 cells/ul, (+): 7-165 cells/ul; (2+): 166-275 cells/ul, (3+): > 275-8000 cells/ul. Qualitative examination of urine WBC: Negative (-): < 10 cells/ul, (+): 11-165 cells/ul; (2+): 166-275 cells/ul, (3+): > 275.

Table 2 Comparison of demographic and clinical features between NCP patients with and without Upro positive

Parameter		Total (N=60)	Upro ⁺ group (n=31)	Upro ⁻ group (n=29)	P ^a
Demographic characteristics					
Gender	Men, n (%)	27(45)	17(28.33)	10(16.67)	0.117
	Women, n (%)	33(55)	14(23.33)	19(31.67)	
Age	(years, IQR)	45(33-54)	49(35-57)	35(26-51)	0.002
Married	Yes, n (%)	50(83.33)	30(96.77)	20(72.41)	0.006
Smoking,	n (%)	3(5)	2(6)	1(3)	0.977
BMI	(Kg/m ² , IQR)	22.9(20.7-25.2)	24(21.7-25.5)	21(20.3-24.0)	0.042
History of hypertension, n (%)		9(15)	8(25.8)	1(3.4)	0.027
History of NCP Contact		60(100)	31(100)	29(100)	1
Disease characteristics					
Temperature (°C, IQR)		36.9(36.7-37.4)	37(36.7-37.7)	37(36.6-37.2)	0.248
Cough, n (%)		32(53.3)	17(54.8)	15(51.7)	0.809
Pharyngalgia, n (%)		10(16.7)	7(22.6)	3(10.3)	0.204
Diarrhoea, n (%)		16(26.7)	8(25.8)	8(27.6)	0.876
Fatigue, n (%)		19, (31.7)	10(32.3)	9(31.0)	0.919
ALT (U/L, IQR)		18(12-27)	21(15-35)	14(11-22)	0.048
AST (U/L, IQR)		24(18-31)	27(23-40)	19(17-25)	0.001
Glb (g/L, IQR)		30(28-33)	32(30-35)	28(26-30)	0.001
Alb (g/L, IQR)		40(39-45)	40(36-45)	41(39-45)	0.168
Scr (μmol/L,IQR)		62(53-72)	68(61-84)	62(53-72)	0.002
BUN (mmol/L,IQR)		4.0(3.1-4.7)	4.5(3.5-5.2)	3.3(2.75-4.15)	0.065
LDH (U/L, IQR)		186(149-284)	223(158-343)	159(141-191)	0.013
CK (U/L, IQR)		63(44-89)	71(51-91)	57(35-78)	0.825
WBC (10 ⁹ /L, IQR)		5.2(4.1-6.4)	4.9(4.5-6.3)	5.0(4.1-7.8)	0.577
NEUT (10 ⁹ /L, IQR)		3.3(2.5-4.7)	3.4(2.9-4.1)	3.0(2.3-5.4)	0.965
LY (10 ⁹ /L, IQR)		1.2(0.9-1.7)	1.1(0.8-1.2)	1.3(1.0-1.9)	0.014
RBC (10 ¹² /L, IQR)		4.6(4.3-4.9)	4.7(4.2-5.0)	4.5(4.4-4.7)	0.5
HB (g/L, IQR)		136(127-147)	136(127-149)	134(127-142)	0.172
CRP (mg/L, IQR)		9(4-17)	17(8-28)	8(3-9)	0.001
Radiographic status^b n (%)					
Normal	(-)	8(13.33)	2(6.45)	6(20.69)	0.215
Mild	(+)	24(40)	10(32.26)	14(48.28)	0.206
Moderate	(2+)	28(4.67)	19(61.29)	9(31.03)	0.037
Severe	(3+)	0(0)	0(0)	0(0)	-

Notes. ^aComparison between the Upro⁺ group and the Upro⁻ group. Data are described as the median (interquartile range,IQR) unless stated otherwise. ^bThe grading criteria as follows: Severe(3+): lobar consolidation; Moderate(2+): grinding glass-like changes; Mild(+): focal infiltration; Normal(-): no infiltration.

Table 3 Comparison of abnormal laboratory indicators between NCP patients with and without Upro positive [n(%)]

Parameter	Total (N=60)	Upro ⁺ group (n=31)	Upro ⁻ group (n=29)	P ^a
ALT	12(26.67)	9(29.03)	3(10.34)	0.071
AST	10(16.67)	9(29.03)	1(6.90)	0.021
Glb	8(13.33)	6(19.35)	2(6.90)	0.156
Alb	8(13.33)	6(19.35)	2(6.90)	0.156
Scr	1(1.67)	1(3.22)	0(0.00)	0.517
BUN	13(21.67)	5(16.13)	8(27.59)	0.282
LDH	16(26.67)	11(35.48)	5(17.24)	0.110
CK	2(3.33)	2(6.45)	0(0.00)	0.263
WBC	16(26.67)	7(22.58)	9(31.03)	0.459
LY	14(23.33)	7(22.58)	7(24.14)	0.887
RBC	18(30.00)	12(38.71)	6(20.69)	0.128
HB	16(26.67)	9(29.03)	7(24.14)	0.668
CRP	9(4-17)	17(8-28)	8(3-9)	0.021

Note. Range of normal value of laboratory indicators: ALT (8 - 40 U/L), ALT (8 - 40 U/L), ALB (35 - 51g/L), ALP(male: 45-125U/L, female: 50-135U/L), SCR (57-111umol/L), BUN (2.9-7.5mmol/L), LDH (≤ 250 U/L), CK (18-198U/L), WBC ($4-10 \times 10^9/L$), CRP (< 8 mg/L) LY ($0.8 \sim 3.5 \times 10^9/L$), RBC (male: $4.5-5.5 \times 10^{12}/L$, female: $4.0-5.0 \times 10^{12}/L$), HGB (male:120-165g/L; female:110-150g/L).

Table 4 Redit analyse for imaging changes in Pulmonary HRCT of the two group

group	n	Imaging changes in Pulmonary HRCT ^a				Ridit value	P value
		Severe(3+)	Moderate(2+)	Mild(+)	Normal(-)		
UPro +	31	0	19	10	2	0.58±0.24	0.012
UPro -	29	0	9	14	6	0.41±0.26	

Note. ^aSevere(3+): lobar consolidation; Moderate(2+): grinding glass-like changes; Mild(+): focal infiltration; Normal(-): no infiltration.